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-: Assignment no. 7: -

No SQL: Design and Develop Queries using CRUD operations. (Use CRUD operations, SAVE method, logical operators, joins etc.).

```
test> use ecommerce
switched to db ecommerce
ecommerce>

ecommerce> db.products.insertMany([
|     { name: "Laptop", category: "electronics", status: "active", quantity: 100, createdAt: new Date() },
|     { name: "T-shirt", category: "clothing", status: "active", quantity: 50, createdAt: new Date() },
|     { name: "Headphones", category: "electronics", status: "active", quantity: 75, createdAt: new Date() },
|     { name: "Jeans", category: "clothing", status: "inactive", quantity: 30, createdAt: new Date() },
|     { name: "Smartphone", category: "electronics", status: "active", quantity: 90, createdAt: new Date() }
| ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69de5cf65a141572a23682d1'),
    '1': ObjectId('69de5cf65a141572a23682d2'),
    '2': ObjectId('69de5cf65a141572a23682d3'),
    '3': ObjectId('69de5cf65a141572a23682d4'),
    '4': ObjectId('69de5cf65a141572a23682d5')
  }
```

1. Find Specific Values: Find all documents where the "status" field is equal to "active."

```
ecommerce>  db.products.find({ status: "active" })
[
  {
    _id: ObjectId('69de5cf65a141572a23682d1'),
    name: 'Laptop',
    category: 'electronics',
    status: 'active',
    quantity: 100,
    createdAt: ISODate('2026-04-14T15:27:50.854Z')
  },
  {
    _id: ObjectId('69de5cf65a141572a23682d2'),
    name: 'T-shirt',
    category: 'clothing',
    status: 'active',
    quantity: 50,
    createdAt: ISODate('2026-04-14T15:27:50.854Z')
  },
  {
    _id: ObjectId('69de5cf65a141572a23682d3'),
    name: 'Headphones',
    category: 'electronics',
    status: 'active',
    quantity: 75,
    createdAt: ISODate('2026-04-14T15:27:50.854Z')
  },
  {
    _id: ObjectId('69de5cf65a141572a23682d5'),
    name: 'Smartphone',
    category: 'electronics',
    status: 'active',
    quantity: 90,
    createdAt: ISODate('2026-04-14T15:27:50.854Z')
  }
]
```

2. Query Criteria (AND condition): Find documents with "status" equal to "active" and "category" equal to "electronics."

```
ecommerce> db.products.find({ status: "active", category: "electronics" })
[
  {
    _id: ObjectId('69de5cf65a141572a23682d1'),
    name: 'Laptop',
    category: 'electronics',
    status: 'active',
    quantity: 100,
    createdAt: ISODate('2026-04-14T15:27:50.854Z')
  },
  {
    _id: ObjectId('69de5cf65a141572a23682d3'),
    name: 'Headphones',
    category: 'electronics',
    status: 'active',
    quantity: 75,
    createdAt: ISODate('2026-04-14T15:27:50.854Z')
  },
  {
    _id: ObjectId('69de5cf65a141572a23682d5'),
    name: 'Smartphone',
    category: 'electronics',
    status: 'active',
    quantity: 90,
    createdAt: ISODate('2026-04-14T15:27:50.854Z')
  }
]
```

3. OR Queries: Find documents where the "status" is "active" or "category" is "clothing."

```
ecommerce> db.products.find({ $or: [{ status: "active" }, { category: "clothing" }] })
[
  {
    _id: ObjectId('69de5cf65a141572a23682d1'),
    name: 'Laptop',
    category: 'electronics',
    status: 'active',
    quantity: 100,
    createdAt: ISODate('2026-04-14T15:27:50.854Z')
  },
  {
    _id: ObjectId('69de5cf65a141572a23682d2'),
    name: 'T-shirt',
    category: 'clothing',
    status: 'active',
    quantity: 50,
    createdAt: ISODate('2026-04-14T15:27:50.854Z')
  },
  {
    _id: ObjectId('69de5cf65a141572a23682d3'),
    name: 'Headphones',
    category: 'electronics',
    status: 'active',
    quantity: 75,
    createdAt: ISODate('2026-04-14T15:27:50.854Z')
  },
  {
    _id: ObjectId('69de5cf65a141572a23682d4'),
    name: 'Jeans',
    category: 'clothing',
    status: 'inactive',
    quantity: 30,
    createdAt: ISODate('2026-04-14T15:27:50.854Z')
  },
  {
    _id: ObjectId('69de5cf65a141572a23682d5'),
    name: 'Smartphone',
    category: 'electronics',
    status: 'active',
    quantity: 90,
    createdAt: ISODate('2026-04-14T15:27:50.854Z')
  }
]
```

4. Conditional Semantic Query: Find documents where the "quantity" is greater than 50.

```
ecommerce> db.products.find({ quantity: { $gt: 50 } })
[
  {
    _id: ObjectId('69de5cf65a141572a23682d1'),
    name: 'Laptop',
    category: 'electronics',
    status: 'active',
    quantity: 100,
    createdAt: ISODate('2026-04-14T15:27:50.854Z')
  },
  {
    _id: ObjectId('69de5cf65a141572a23682d3'),
    name: 'Headphones',
    category: 'electronics',
    status: 'active',
    quantity: 75,
    createdAt: ISODate('2026-04-14T15:27:50.854Z')
  },
  {
    _id: ObjectId('69de5cf65a141572a23682d5'),
    name: 'Smartphone',
    category: 'electronics',
    status: 'active',
    quantity: 90,
    createdAt: ISODate('2026-04-14T15:27:50.854Z')
  }
]
```

5. $where Query: Find documents where the value of "field1" plus "field2" is equal to 10.

```
ecommerce> db.products.find({ $where: "this.quantity + 10 === 100" })
[
  {
    _id: ObjectId('69de5cf65a141572a23682d5'),
    name: 'Smartphone',
    category: 'electronics',
    status: 'active',
    quantity: 90,
    createdAt: ISODate('2026-04-14T15:27:50.854Z')
  }
]
```

6. Cursors with Limit and Skip: Get the first 5 documents where "status" is "active," skipping the first 2.

```
ecommerce> db.products.find({ status: "active" }).skip(1).limit(3)
[
  {
    _id: ObjectId('69de5cf65a141572a23682d2'),
    name: 'T-shirt',
    category: 'clothing',
    status: 'active',
    quantity: 50,
    createdAt: ISODate('2026-04-14T15:27:50.854Z')
  },
  {
    _id: ObjectId('69de5cf65a141572a23682d3'),
    name: 'Headphones',
    category: 'electronics',
    status: 'active',
    quantity: 75,
    createdAt: ISODate('2026-04-14T15:27:50.854Z')
  },
  {
    _id: ObjectId('69de5cf65a141572a23682d5'),
    name: 'Smartphone',
    category: 'electronics',
    status: 'active',
    quantity: 90,
    createdAt: ISODate('2026-04-14T15:27:50.854Z')
  }
]
```

7. Cursors with Sort: Find all documents sorted by "createdAt" in descending order.

```
ecommerce> db.products.find().sort({ createdAt: -1 })
[
  {
    _id: ObjectId('69de5cf65a141572a23682d1'),
    name: 'Laptop',
    category: 'electronics',
    status: 'active',
    quantity: 100,
    createdAt: ISODate('2026-04-14T15:27:50.854Z')
  },
  {
    _id: ObjectId('69de5cf65a141572a23682d2'),
    name: 'T-shirt',
    category: 'clothing',
    status: 'active',
    quantity: 50,
    createdAt: ISODate('2026-04-14T15:27:50.854Z')
  },
  {
    _id: ObjectId('69de5cf65a141572a23682d3'),
    name: 'Headphones',
    category: 'electronics',
    status: 'active',
    quantity: 75,
    createdAt: ISODate('2026-04-14T15:27:50.854Z')
  },
  {
    _id: ObjectId('69de5cf65a141572a23682d4'),
    name: 'Jeans',
    category: 'clothing',
    status: 'inactive',
    quantity: 30,
    createdAt: ISODate('2026-04-14T15:27:50.854Z')
  },
  {
    _id: ObjectId('69de5cf65a141572a23682d5'),
    name: 'Smartphone',
    category: 'electronics',
    status: 'active',
    quantity: 90,
    createdAt: ISODate('2026-04-14T15:27:50.854Z')
  }
```

8. Database Commands: Use the db. Stats () command to retrieve statistics about the database.

```
ecommerce> db.runCommand({ dbStats: 1 })
{
  db: 'ecommerce',
  collections: Long('1'),
  views: Long('0'),
  objects: Long('5'),
  avgObjSize: 117.8,
  dataSize: 589,
  storageSize: 20480,
  indexes: Long('1'),
  indexSize: 20480,
  totalSize: 40960,
  scaleFactor: Long('1'),
  fsUsedSize: 12877942784,
  fsTotalSize: 45087715328,
  ok: 1
}
```

VIEW Commands:-

◆ Limit Results

```
ecommerce> db.products.find().limit(3)
[
  {
    _id: ObjectId('69de5cf65a141572a23682d1'),
    name: 'Laptop',
    category: 'electronics',
    status: 'active',
    quantity: 100,
    createdAt: ISODate('2026-04-14T15:27:50.854Z')
  },
  {
    _id: ObjectId('69de5cf65a141572a23682d2'),
    name: 'T-shirt',
    category: 'clothing',
    status: 'active',
    quantity: 50,
    createdAt: ISODate('2026-04-14T15:27:50.854Z')
  },
  {
    _id: ObjectId('69de5cf65a141572a23682d3'),
    name: 'Headphones',
    category: 'electronics',
    status: 'active',
    quantity: 75,
    createdAt: ISODate('2026-04-14T15:27:50.854Z')
  }
]
```

◆ Select Specific Fields

```
ecommerce> db.products.find({}, { name: 1, category: 1 })
[
  {
    _id: ObjectId('69de5cf65a141572a23682d1'),
    name: 'Laptop',
    category: 'electronics'
  },
  {
    _id: ObjectId('69de5cf65a141572a23682d2'),
    name: 'T-shirt',
    category: 'clothing'
  },
  {
    _id: ObjectId('69de5cf65a141572a23682d3'),
    name: 'Headphones',
    category: 'electronics'
  },
  {
    _id: ObjectId('69de5cf65a141572a23682d4'),
    name: 'Jeans',
    category: 'clothing'
  },
  {
    _id: ObjectId('69de5cf65a141572a23682d5'),
    name: 'Smartphone',
    category: 'electronics'
  }
]
```